

EE6407 Genetic Algorithms and Machine Learning

Quiz 1 Preparation Notes: Evolutionary Computing Part 1 (Weeks 1–4)

Living course note

24 August 2026

Scope and sources. This is the detailed, self-contained teaching note for Quiz 1. It covers the complete four-week Part 1 scope: problem classification, computational complexity, evolutionary-computing foundations, the evolutionary-algorithm loop and components, 8-queens, the simple genetic algorithm, and optimisation context. The authoritative source is `../resources/lecture-01-evolutionary-computing-part-1.pdf`, PDF pp. 1–36, supported by the Week 1 and Week 2 transcripts. Historical questions are used only as pattern evidence; every worked answer here is independently derived.

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1 Evolutionary computing in one idea

Evolutionary computing (EC) is a family of methods for searching for good solutions by maintaining and improving a population of candidates. Genetic algorithms (GAs) are an important member of this family. In lecture, the names *evolutionary computing*, *evolutionary algorithm*, and *genetic algorithm* may be used informally, but strictly a GA is one specific type of evolutionary algorithm.

EC is useful when a problem has a large, irregular, discrete, mixed, or non-differentiable search space, or when exact optimisation would be too expensive. It does not automatically guarantee the best possible answer. It is a stochastic method for finding high-quality candidates within a computational budget.

2 Classifying the problem before choosing a method

Represent a system abstractly as $y = M(x)$:

- If M and desired output y are known but x is unknown, we have an **optimisation** problem.
- If input–output pairs are known but M is unknown, we have a **modelling** or learning problem.
- If x and M are known but y is unknown, we have a **simulation** problem.

Many machine-learning tasks can be written as optimisation: select model parameters to minimise prediction error. The distinction is still useful because it clarifies what information is available and what the method must find.

The *search space* S contains all candidates the algorithm may consider. The *feasible set* $\mathcal{F} \subseteq S$ contains those satisfying every hard constraint. An objective function $f : S \rightarrow \mathbb{R}$ measures quality. A constraint-satisfaction problem asks for any member of $\mathcal{F}$; a constrained optimisation problem asks for the best member.

3 Why search can become difficult

The number of candidate solutions can grow much faster than intuition suggests. For a route through n labelled cities, there are $n!$ possible orders before considering symmetries. Exhaustive enumeration is therefore impractical even at modest n .

Complexity words provide context. A decision problem is in **P** if it can be solved in polynomial time. It is in **NP** if a proposed yes-answer can be verified in polynomial time. NP-complete problems are the hardest known problems in NP under polynomial reductions, and NP-hard problems are at least as hard but need not be decision problems. Whether **P** = **NP** is unresolved. These labels motivate heuristic search; they do not prove that an EA will solve a particular instance well.

4 The simplified evolutionary metaphor

Evolutionary algorithms borrow four ideas from a simplified account of natural evolution: resources are limited, individuals differ, some differences affect reproductive success and are heritable, and random variation supplies new differences. In computing, *fitness* is a deliberately designed numerical score rather than a biological fact.

The terms below connect the problem to its representation:

- A *phenotype* is the candidate solution in the original problem domain.
- A *genotype* is the encoding manipulated by the algorithm.
- A *chromosome* is a complete genotype; a *gene* is one component; an *allele* is a possible value of that component.

The representation is an engineering decision. It should make valid solutions reachable and should work naturally with the variation operators introduced later. A poor representation can make a simple problem unnecessarily difficult.

5 Foundations checkpoint

- Can I distinguish optimisation, modelling, and simulation using $y = M(x)$?
- Can I identify a search space, a feasible set, and an objective?
- Can I explain why combinatorial spaces make exhaustive search difficult?
- Can I distinguish evolutionary computing from a genetic algorithm?
- Can I explain genotype, phenotype, chromosome, gene, and allele?

Source: `../resources/lecture-01-evolutionary-computing-part-1.pdf`, PDF pp. 2–16;
`../resources/week-01-transcript.txt`.

6 Quiz 1 scope and evidence

6.1 What is confirmed, and what is only supporting evidence?

This note prepares the complete four-week Part 1 scope presently contained in `../resources/lecture-01-evolutionary-computing-part-1.pdf`. The supplied deck and transcript establish a closed-book written Quiz 1 worth 10%, held in the final hour of Week 4. The exact date, time, and venue must still be checked in NTULearn; tentative transcript details are not treated as confirmed.

Confidence	Evidence	How it is used here
Current and authoritative	Lecture 1 Part 1 deck, PDF pages 1–36	Defines the examinable Part 1 concepts: classification, complexity, EC foundations, the EA loop and components, 8-queens, the simple GA example, EA behaviour, and optimisation context.
Current supporting context	Week 1 and Week 2 transcripts	Supports the assessment format and lecturer emphasis. Any tentative logistics remain unconfirmed.
Historical pattern only	Legacy quiz PDF pages 1–2; 2021, 2022, and 2024 walkthroughs	Suggests useful question forms involving classification, representation, encoding, fitness, and a hand-worked SGA generation. Answers below are newly derived.
Frequency evidence only	Exam-topic workbook, Sheet1 A1:G24	Shows recurring older-exam themes. It does not establish the present Quiz 1 scope.

Deliberately excluded. Differential evolution and particle swarm optimisation are outside this preparation pack. Also excluded are advanced crossover families, schema theory, niching, multi-objective optimisation, and machine learning. These topics appeared in older or later materials but are not in the approved Part 1 scope.

Source: Lecture 1 Part 1 deck, PDF pp. 1–36; Week 1 and Week 2 transcripts; historical-pattern sources indexed in `../STATUS.md`.

7 One-page memory core

Problem classification	For $y = M(x)$: unknown x means optimisation; unknown M means modelling/learning; unknown y means simulation.
Search language	Search space S : all encodable candidates. Feasible set $\mathcal{F} \subseteq S$: candidates satisfying hard constraints. Objective f : quality. CSP seeks any feasible candidate; COP seeks the best feasible candidate.
Complexity	P : solvable in polynomial time. NP : a proposed yes-certificate is verifiable in polynomial time. NP-complete: in NP and NP-hard. NP-hard: at least as hard as every NP problem, but may be an optimisation problem and need not be in NP.
EA cycle	Initialise $\rightarrow$ evaluate $\rightarrow$ select parents $\rightarrow$ vary by recombination/mutation $\rightarrow$ evaluate offspring $\rightarrow$ select survivors $\rightarrow$ test termination.
Representation	Phenotype is the domain solution; genotype is its encoding. A useful encoding makes intended solutions reachable and lets operators preserve or repair validity.
Selection	Fitness-proportionate selection: $p_i = f_i / \sum_j f_j$ when fitnesses are non-negative and larger is better. Selection favours; it does not guarantee.
8-queens	Permutation $q = (q_1, \dots, q_N)$ places one queen in column i , row q_i . Queens i, j conflict diagonally when $ q_i - q_j = i - j $. With C diagonal conflicts, $f = \binom{N}{2} - C$.
Binary decoding	For bits $b_{L-1} \dots b_0$, $z = \sum_{k=0}^{L-1} 2^k b_k$. Optional interval map: $x = a + \frac{z}{2^L - 1}(b - a)$.
Mutation checks	Across N strings of length L , expected flipped bits are $N L p_m$. Probability of no flip is $(1 - p_m)^{NL}$.
Optimisation context	A neighbourhood defines local moves. A local optimum is best only in its neighbourhood; a global optimum is best in the whole feasible set. No single optimiser is best across all possible problems.

Closed-book recall test. Cover the table and reconstruct the three $y = M(x)$ cases, the P/NP relationships, the seven EA stages, the 8-queens conflict test, roulette selection, and binary decoding.

8 Five worked archetypes

8.1 Archetype 1: classify the problem before choosing an algorithm

Question. A factory knows a simulator M that predicts energy use y for a machine setting x . Classify each task:

- (a) Find a setting that minimises energy while meeting production constraints.
- (b) Predict the energy used by one specified setting.
- (c) Infer a predictive relationship from logged settings and measured energy.

Construction. Write $y = M(x)$ and ask which object is unknown.

- (a) M is known and the desired low-energy outcome is specified, but x is unknown: **optimisation**. The search space contains encodable settings, the feasible set satisfies production constraints, and the objective is predicted energy.
- (b) x and M are known, while y is requested: **simulation**.
- (c) Input–output examples are known, while M is unknown: **modelling/learning**.

Trap. The use of a computer model does not automatically make a task simulation. Classification depends on what is unknown.

8.2 Archetype 2: reason correctly about P, NP, NP-complete, and NP-hard

Question. Assess the claim that NP means non-polynomial and therefore an NP-hard optimisation problem cannot be checked efficiently.

Answer. The claim combines two errors.

1. NP means *nondeterministic polynomial time*, commonly characterised by polynomial-time verification of a proposed yes-certificate for a decision problem. It does not mean that the problem is proved to require non-polynomial time.
2. NP-hard is a hardness relation. An NP-hard problem need not itself be a decision problem or belong to NP. For an optimisation problem, checking that a candidate is feasible and calculating its objective may be easy even when proving that it is globally optimal is hard.

If a decision problem is both in NP and NP-hard, it is NP-complete. Every problem in P is also in NP. Whether $\mathbf{P} = \mathbf{NP}$ remains unresolved.

8.3 Archetype 3: trace the evolutionary algorithm loop

Question. A population contains four binary strings. Describe one generation without confusing parent selection and survivor selection.

1. **Represent and initialise:** choose an encoding and create the initial population.
2. **Evaluate:** decode each genotype, calculate its objective, and convert it to the fitness used by selection.
3. **Select parents:** sample candidates for reproduction. Better fitness normally means greater probability, not certainty.
4. **Recombine:** exchange genetic material between selected parents to make offspring.
5. **Mutate:** randomly change genes, supplying new variation.

6. **Evaluate offspring:** decode and score the new candidates.
7. **Select survivors:** decide which parents and/or offspring form the next population.
8. **Terminate or repeat:** stop at a target, budget, or stagnation condition; otherwise begin the next generation.

Trap. Parent selection answers who reproduces; survivor selection answers who remains. They can use different rules.

8.4 Archetype 4: represent and score 8-queens

For $N = 4$, let q_i be the row of the queen in column i . The permutation $q = (2, 4, 1, 3)$ automatically gives one queen in each row and column. For every pair $i < j$, compare $|q_i - q_j|$ with $|i - j|$:

(i, j)	(1, 2)	(1, 3)	(1, 4)	(2, 3)	(2, 4)	(3, 4)
$ q_i - q_j $	2	1	1	3	1	2
$ i - j $	1	2	3	1	2	1

No pair is equal, so $C = 0$ and $f(q) = \binom{4}{2} - C = 6$. A swap mutation exchanging the last two positions gives $q' = (2, 4, 3, 1)$. It remains a valid permutation, but pair (2, 3) conflicts, so $C = 1$ and $f(q') = 5$.

Operator lesson. Swap mutation preserves the permutation. Recombination must likewise preserve it or be followed by repair.

8.5 Archetype 5: calculate one complete SGA generation

Maximise $f(x) = x^2$ for an unsigned four-bit integer. Start with:

chromosome	x	$f(x)$	$p_i = f_i / \sum f_j$
0011	3	9	$9/311 = 0.0289$
0101	5	25	$25/311 = 0.0804$
1001	9	81	$81/311 = 0.2605$
1110	14	196	$196/311 = 0.6302$

The fitness sum is 311. Suppose roulette draws produce the mating pool [1110, 1001, 1110, 0101]. Pair adjacent parents and cross after bit 2:

$$\begin{aligned} 11|10, 10|01 &\longrightarrow 11|01 = 1101, & 10|10 &= 1010, \\ 11|10, 01|01 &\longrightarrow 11|01 = 1101, & 01|10 &= 0110. \end{aligned}$$

Suppose mutation flips the final bit of the fourth offspring only, so $0110 \rightarrow 0111$. With generational replacement the next population is [1101, 1010, 1101, 0111], decoding to [13, 10, 13, 7], with fitnesses [169, 100, 169, 49]. Its total fitness is 487 and its best fitness is 169.

Checks. The roulette outcomes and mutation event are sampled events, not guarantees. Decode after variation, recompute fitness, and use the correct generation's fitness sum.

9 Progressive drills

Attempt all twelve without notes. Hints are brief; complete solutions begin on the next page.

- D1.** For $y = M(x)$, classify: (i) estimating M from examples; (ii) finding x for a desired y with known M ; (iii) finding y for known x, M .

Hint: identify the unknown object.

- D2.** A delivery problem has all possible route encodings S , a maximum-distance rule, and total fuel use f . Identify the search space, feasible set, and objective.

Hint: constraints define a subset of the search space.

- D3.** How many ordered routes are there through eight labelled cities before symmetry reductions? Why does this number matter?

Hint: order all eight objects.

- D4.** Correct the claim that every NP-hard problem is an NP-complete decision problem and NP means non-polynomial.

Hint: separate membership from hardness.

- D5.** Put these in a valid EA order: mutation, termination test, initialisation, offspring evaluation, parent selection, survivor selection, population evaluation, recombination.

Hint: candidates must exist before evaluation.

- D6.** Three candidates have fitnesses 2, 3, 5. Calculate their roulette-selection probabilities.

Hint: divide by total fitness.

- D7.** Explain in one sentence why repeatedly selecting only the current best candidate can be risky.

Hint: think about diversity and local optima.

- D8.** For $q = (2, 4, 1, 3)$ in 4-queens, calculate the diagonal conflicts and fitness $\binom{4}{2} - C$.

Hint: test every pair.

- D9.** Parents (1, 2, 3, 4) and (4, 3, 2, 1) undergo ordinary one-point crossover after position 2. Write the children and explain the validity problem.

Hint: a permutation contains each row once.

- D10.** Decode the unsigned binary string 10110.

Hint: use place values 16, 8, 4, 2, 1.

- D11.** Map 10110 linearly to $[10, 20]$, including both endpoints.

Hint: use $10 + \frac{z}{31}(20 - 10)$.

- D12.** Six strings each have length eight and independent bit-mutation probability $p_m = 0.005$. Find the expected flips and probability of no flips.

Hint: there are 48 trials.

10 Solutions to the progressive drills

D1. (i) Modelling/learning; (ii) optimisation; (iii) simulation.

D2. S is every encodable route; $\mathcal{F}$ is the subset obeying the maximum-distance rule; minimise total fuel use f .

D3. $8! = 40320$. Factorial growth makes exhaustive enumeration rapidly expensive.

D4. NP is polynomial-time verification of yes-certificates for decision problems, not non-polynomial. NP-hard problems need not be in NP or be decision problems. NP-complete problems are both in NP and NP-hard.

D5.

initialise $\rightarrow$ evaluate population $\rightarrow$ select parents $\rightarrow$ recombine $\rightarrow$ mutate
 $\rightarrow$ evaluate offspring $\rightarrow$ select survivors $\rightarrow$ test termination.

D6. Total 10; probabilities 0.2, 0.3, 0.5, summing to 1.

D7. It can destroy diversity and cause premature convergence around a local optimum.

D8. No conflicting pair, so $C = 0$ and $f = 6$.

D9. Children $(1, 2, 2, 1)$ and $(4, 3, 3, 4)$ repeat and omit rows; use a permutation-preserving crossover or repair.

D10. $10110_2 = 16 + 4 + 2 = 22$.

D11. $x = 10 + \frac{22}{31}(10) = 17.0968$ approximately.

D12. Expected flips $= 6(8)(0.005) = 0.24$; no-flip probability $= 0.995^{48} \approx 0.7862$.

11 Mock Quiz A: foundations and a hand-worked generation

Conditions: 60 minutes, closed book, 100 marks.

Question 1: Problem formulation (20 marks; 10 minutes)

A trained surrogate predicts drag $y = M(x)$ from an airfoil design x .

- (a) Classify finding the lowest-drag design subject to strength constraints. Identify S , $\mathcal{F}$, and the objective. [10]
- (b) Classify (i) predicting drag for a given design and (ii) learning the surrogate from examples. [6]
- (c) Distinguish a CSP from a constrained optimisation problem. [4]

Question 2: Complexity and EC foundations (20 marks; 10 minutes)

- (a) Define P, NP, NP-complete, and NP-hard. [12]
- (b) Explain why NP-hardness motivates heuristic search but does not prove an EA will perform well. [4]
- (c) Distinguish genotype from phenotype, with one example. [4]

Question 3: The EA loop (20 marks; 10 minutes)

- (a) Write one complete generation of a general EA in a correct order. [12]
- (b) Distinguish parent selection from survivor selection. [4]
- (c) Give two sensible termination conditions. [4]

Question 4: 4-queens representation (20 marks; 12 minutes)

Use q_i = row in column i and fitness $f = \binom{4}{2} - C$.

- (a) For $q = (1, 3, 4, 2)$, list every diagonal conflict and calculate f . [10]
- (b) Explain which constraints the permutation enforces automatically. [4]
- (c) Explain why ordinary one-point crossover may fail and give a remedy. [6]

Question 5: Simple GA calculation (20 marks; 18 minutes)

Maximise $f(x) = x^2$ using three-bit unsigned strings. The population is 010, 011, 101, 110.

- (a) Decode, calculate all fitnesses, and calculate roulette probabilities. [10]
- (b) The mating pool is 110, 101, 011, 110. Cross adjacent parents after bit 1. [6]
- (c) Flip the last bit of the first child only. Give the next population and best fitness. [4]

12 Mock Quiz A: marking guide

Q1 (20). Optimisation (2); S all encodable designs (2); $\mathcal{F}$ strength-feasible designs (2); minimise drag (2); known M /unknown x (2). Simulation (3), modelling/learning (3). CSP seeks any feasible candidate (2); constrained optimisation seeks the best (2).

Q2 (20). P solvable in polynomial time (3); NP has polynomially verifiable yes-certificates (3); NP-complete is in NP and NP-hard (3); NP-hard need not be in NP (3). Motivation without algorithm-specific guarantee (4). Genotype is encoding; phenotype is domain solution, with example (4).

Q3 (20). Initialise, evaluate, parent select, recombine, mutate, evaluate offspring, survivor select, terminate/repeat: 12. Parent selection chooses reproducers; survivor selection forms the next population: 4. Any two of target, evaluation/time budget, or stagnation: 4.

Q4 (20). Only (2,3) conflicts (6); $C = 1$, $f = 5$ (4). One queen per column and row (4). Ordinary crossover can duplicate/omit rows (3); permutation-preserving crossover or repair (3).

Q5 (20). Decodings 2, 3, 5, 6 (2); fitnesses 4, 9, 25, 36 (2); sum 74 (2); probabilities $4/74$, $9/74$, $25/74$, $36/74$ (4). The crosses are

$$1|10, 1|01 \rightarrow 101, 110 \quad \text{and} \quad 0|11, 1|10 \rightarrow 010, 111$$

(6). Mutate $101 \rightarrow 100$; the next population is 100, 110, 010, 111 (2), and its best fitness is 49 (2).

Total: 100 marks.

13 Mock Quiz B: representation, operators, and optimisation context

Conditions: 60 minutes, closed book, 100 marks.

Question 1: Representation and encoding (20 marks; 12 minutes)

- (a) Define chromosome, gene, allele, genotype, and phenotype. [10]
- (b) Decode 11010_2 and map it linearly to $[-2, 3]$, including endpoints. [6]
- (c) State two properties of a useful representation. [4]

Question 2: Selection and variation (20 marks; 10 minutes)

Four candidates have fitnesses 1, 2, 3, 4.

- (a) Calculate roulette probabilities. [6]
- (b) Explain why the best candidate is not guaranteed to be selected. [4]
- (c) Distinguish recombination from mutation and state one role of each. [6]
- (d) State one danger of excessive selection pressure. [4]

Question 3: 5-queens fitness (20 marks; 12 minutes)

Use $q = (1, 2, 3, 5, 4)$ and fitness $f = \binom{5}{2} - C$.

- (a) Identify all diagonal conflicts. [12]
- (b) Calculate C and f . [4]
- (c) Explain why swap mutation preserves row/column feasibility. [4]

Question 4: SGA checks (25 marks; 16 minutes)

Five strings have length six; bit mutation is independent with $p_m = 0.01$.

- (a) Find expected flips and probability of no flips. [8]
- (b) Cross $101|011$ and $010|110$ at the shown point. [5]
- (c) Decode both children as unsigned integers. [4]
- (d) For minimising $g(x) = (x - 20)^2$, explain why raw g is wrong in a larger-is-better roulette rule and give a valid transformation. [8]

Question 5: Search behaviour (15 marks; 10 minutes)

- (a) Distinguish local and global optima using a neighbourhood. [6]
- (b) Explain exploration versus exploitation. [5]
- (c) State the practical message of no free lunch. [4]

14 Mock Quiz B: marking guide

Q1 (20). Chromosome/genotype, gene/component, allele/value, genotype/encoding, phenotype/domain solution: 2 each. $11010_2 = 26$ (2); $x = -2 + \frac{26}{31}(5) = 2.1935$ (4). Any two useful representation properties: 2 each.

Q2 (20). Total 10, probabilities 0.1, 0.2, 0.3, 0.4 (6). Stochastic selection is not certainty (4). Recombination combines parental material (3); mutation supplies random new variation (3). Excess pressure can destroy diversity and cause premature convergence (4).

Q3 (20). Conflicts (1, 2), (1, 3), (2, 3), (4, 5): 3 each. Thus $C = 4$, $f = 10 - 4 = 6$ (4). Swapping retains each row once, while columns are positions (4).

Q4 (25). There are 30 bits; expected flips 0.3 (4); no-flip probability $0.99^{30} \approx 0.7397$ (4). Children 101110 and 010011 (5), decoding to 46 and 19 (4). Raw g rewards larger error (4); for example $F = 1/(1 + g)$ is positive and decreases with error (4).

Q5 (15). Local is best in its neighbourhood; global is best in the whole feasible set (6). Exploration samples new regions; exploitation improves promising regions (5). No optimiser dominates all possible problems, so problem knowledge and empirical testing matter (4).

Total: 100 marks.

15 Eight-day preparation schedule: 24–31 August

Date	Focus	Minimum deliverable
24 Aug	Scope and classification	Reproduce the scope map; solve D1–D3 without notes.
25 Aug	Complexity and EC foundations	Explain P/NP/NP-complete/NP-hard aloud; solve D4.
26 Aug	EA loop and components	Draw the complete loop from memory; solve D5–D7.
27 Aug	Representation and 8-queens	Recalculate Archetype 4; solve D8–D9.
28 Aug	SGA arithmetic	Recalculate Archetype 5; solve D10–D12 and check every sum.
29 Aug	Timed Mock A	Attempt in 60 minutes; mark in a different colour; relearn every lost mark.
30 Aug	Timed Mock B	Attempt in 60 minutes; mark and write a one-line cause for each error.
31 Aug	Retrieval and weak spots	Recite the memory core, redo missed questions, and stop heavy study early.

15.1 Quiz-day checklist

- Verify the official date, time, venue, and permitted materials in NTU Learn.
- Before calculating, write the encoding, objective direction, and fitness convention.
- For roulette selection, calculate the fitness sum and check probabilities sum to 1.
- After crossover or mutation, check chromosome length and representation validity.
- For 8-queens, test each pair once using $i < j$; do not double-count.
- For complexity questions, do not expand NP as non-polynomial.
- Reserve the final five minutes for arithmetic and unanswered subparts.

Source: ../resources/lecture-01-evolutionary-computing-part-1.pdf, PDF pp. 1–36. Historical sources were used only to design original question patterns; all calculations and solutions here were independently derived..